CODE:

```
from sklearn.feature_extraction.text import TfidfVectorizer
import pandas as pd

documents = [
    "I love machine learning",
    "I love python programming",
    "Machine learning is interesting"
]

vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform(documents)

words = vectorizer.get_feature_names_out()

df = pd.DataFrame(
    tfidf_matrix.toarray(),
    columns=words,
    index=["Document 1", "Document 2", "Document 3"]
)

print("TF-IDF Values:\n")
print(df.round(4).to_string())
```

OUTPUT:

```
PS C:\Users\kimax\AppData\Local\Programs\Microsoft VS Code> & 'c:\python314\python.exe' 'c:/Users/kimax/Documents/JU/Sem-3/NLP/Activity
-2/main.py'
TF-IDF Values:

            interesting      is  learning    love  machine  programming  python
Document 1       0.0000  0.0000    0.5774  0.5774   0.5774       0.0000  0.0000
Document 2       0.0000  0.0000    0.0000  0.4736   0.0000       0.6228  0.6228
Document 3       0.5628  0.5628    0.4280  0.0000   0.4280       0.0000  0.0000
PS C:\Users\kimax\AppData\Local\Programs\Microsoft VS Code>
```